

The Binomial Theorem (7.4)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 45 # _____ How many terms are in the expansion of $(a + b)^5$?	ANSWER: 1 # _____ Evaluate $\binom{8}{2}$.
ANSWER: 21 # _____ Write Row 4 of Pascal's Triangle.	ANSWER: 36 # _____ Find the 3rd term of the expansion of $(x + 2)^3$.

ANSWER: 6

_____ Evaluate $\binom{7}{2}$.

ANSWER: 10

_____ Evaluate $\binom{9}{2}$.

ANSWER: 20

_____ What are the first and last coefficients in any expansion of $(a + b)^n$?

ANSWER: $12x$

_____ Evaluate $\binom{10}{2}$.

ANSWER: 1, 4, 6, 4, 1

_____ Evaluate $\binom{6}{3}$.

ANSWER: 28

_____ How many terms are in the expansion of $(a + b)^9$?